

QVerse — Week 2: The Minimum Mathematics You Need

Theory Booklet | Introduction to Quantum Computing & Programming

QVerse

2026

How this booklet fits the course

This theory booklet is designed to be read alongside `02_minimum_quantum_math.ipynb`. It contains the conceptual and mathematical material that should be explained during Week 2; the notebook then turns those ideas into experiments, circuit construction, and exercises.

Learning objectives

- Work with complex amplitudes.
- Normalize state vectors.
- Convert amplitudes into measurement probabilities.
- Interpret a gate as a matrix acting on a vector.

Recommended workflow

1. Read the relevant theory section.
2. Predict the result of the corresponding notebook experiment.
3. Run the notebook.
4. Explain any difference between the prediction and the output.
5. Complete the core exercises without treating the solution notebook as a substitute for reasoning.

1. Why mathematics enters quantum computing

Quantum computation is built on linear algebra. A qubit is described by a vector, a gate by a matrix, and measurement probabilities by the squared magnitudes of complex amplitudes.

The goal of this week is to introduce only the mathematics that is needed immediately. The full machinery of quantum mechanics is not required.

The four core ideas are:

1. amplitudes can be complex numbers;
2. a valid quantum state must be normalized;
3. probabilities are obtained from amplitude magnitudes;
4. gates act by matrix-vector multiplication.

2. Complex numbers

A complex number has the form

$$z = a + ib,$$

where $a, b \in \mathbb{R}$ and $i^2 = -1$.

Its complex conjugate is

$$z^* = a - ib,$$

and its magnitude is

$$|z| = \sqrt{z^* z} = \sqrt{a^2 + b^2}.$$

Therefore,

$$|z|^2 = a^2 + b^2.$$

The squared magnitude is especially important because quantum probabilities are computed from $|\alpha|^2$, not directly from α .

Two complex amplitudes can have the same magnitude but different phase. For example,

$$\frac{1}{\sqrt{2}} \quad \text{and} \quad \frac{i}{\sqrt{2}}$$

have identical magnitude $1/\sqrt{2}$, but different complex phase. That phase difference can affect later interference even though the immediate measurement probabilities may be the same.

3. Qubit state vectors

A general pure one-qubit state is

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,$$

where α and β are complex amplitudes.

Using column-vector notation,

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

so

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

The two descriptions are equivalent. Bra-ket notation is usually more readable for algorithms; vectors and matrices are useful for calculations.

4. Normalization

A quantum state must satisfy

$$|\alpha|^2 + |\beta|^2 = 1.$$

This is the normalization condition.

Why? Because the total probability of observing one of the possible computational-basis outcomes must be 1:

$$P(0) + P(1) = 1.$$

For a candidate vector

$$v = \begin{pmatrix} a \\ b \end{pmatrix},$$

its norm is

$$\|v\| = \sqrt{|a|^2 + |b|^2}.$$

If $v \neq 0$, the normalized state is

$$|\psi\rangle = \frac{v}{\|v\|}.$$

For example, $[1, 1]^T$ is not normalized because

$$|1|^2 + |1|^2 = 2.$$

Dividing by $\sqrt{2}$ gives

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

5. The Born rule

For

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,$$

measurement in the computational basis gives

$$P(0) = |\alpha|^2, \quad P(1) = |\beta|^2.$$

This is a simple form of the **Born rule**.

The amplitudes themselves are not probabilities. They may be negative or complex. The probabilities are always real and non-negative because they are squared magnitudes.

Example:

$$|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle.$$

Then

$$P(0) = \frac{3}{4}, \quad P(1) = \frac{1}{4}.$$

6. Bras, inner products, and orthogonality

If

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix},$$

the corresponding bra is the conjugate transpose

$$\langle\psi| = (\alpha^* \quad \beta^*).$$

The inner product of two states is

$$\langle\phi|\psi\rangle.$$

A normalized state satisfies

$$\langle\psi|\psi\rangle = 1.$$

The computational basis states are orthogonal:

$$\langle 0|1\rangle = 0.$$

Orthogonality is what allows the two basis states to represent perfectly distinguishable measurement outcomes.

7. Gates as matrices

A one-qubit gate is represented by a 2×2 unitary matrix U . It acts by matrix-vector multiplication:

$$|\psi'\rangle = U|\psi\rangle.$$

For example,

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Then

$$X|0\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle.$$

The same matrix also acts on superpositions. This is why a quantum gate cannot be fully characterized by a classical truth table.

8. Why quantum gates are unitary

A matrix U is unitary if

$$U^\dagger U = U U^\dagger = I,$$

where $U^\dagger$ is the conjugate transpose.

Unitarity preserves inner products and therefore preserves normalization:

$$\langle \psi' | \psi' \rangle = \langle \psi | U^\dagger U | \psi \rangle = \langle \psi | \psi \rangle.$$

This is the mathematical reason ideal closed-system quantum gates are reversible.

If U is unitary, its inverse is

$$U^{-1} = U^\dagger.$$

Measurement is different: it is not represented as an ordinary reversible unitary gate in this simple circuit model.

9. The Hadamard matrix

The Hadamard gate is

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Applying it to the computational basis gives

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}},$$

and

$$H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

These states will be called $|+\rangle$ and $|-\rangle$ in Week 3.

A crucial identity is

$$H^2 = I.$$

Therefore the Hadamard gate is its own inverse.

10. Matrix multiplication order

If a state first experiences U_1 and then U_2 , the final state is

$$|\psi_f\rangle = U_2 U_1 |\psi_0\rangle.$$

The rightmost matrix acts first.

This ordering often causes mistakes when students translate a circuit into algebra. A circuit is read left to right, while the corresponding matrix product is written right to left.

11. Global phase

Two statevectors that differ only by an overall phase

$$|\psi'\rangle = e^{i\gamma} |\psi\rangle$$

represent the same physical pure state for the measurements considered in this course.

For example,

$$|0\rangle \quad \text{and} \quad -|0\rangle$$

have the same measurement statistics and differ only by global phase.

In contrast, changing the phase of only one component can create a **relative phase**, which can affect interference. This distinction becomes central in Weeks 3 and 4.

12. Why NumPy is useful

The notebook uses NumPy because it makes the mathematical objects explicit:

- complex vectors represent amplitudes;
- arrays represent gates;
- matrix multiplication verifies identities;
- helper functions can normalize arbitrary vectors.

This is not “extra programming.” It provides an independent way to check what Qiskit is doing.

A strong workflow is:

1. predict the result algebraically;
2. verify it with NumPy;
3. verify it again with Qiskit’s statevector tools.

Agreement among the three views is a useful debugging check.

13. Common mistakes

Confusing amplitude and probability

$$\alpha \neq P(0), \quad |\alpha|^2 = P(0).$$

Forgetting complex conjugation

The norm uses z^*z , not z^2 .

Normalizing by the sum instead of the norm

The correct factor is

$$\sqrt{\sum_j |a_j|^2}.$$

Treating every matrix as a valid quantum gate

Ideal gate matrices must be unitary.

Thinking a negative amplitude means negative probability

Probabilities are squared magnitudes and are never negative.

14. What to know before opening the notebook

You should be comfortable with:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,$$

$$|\alpha|^2 + |\beta|^2 = 1,$$

$$P(0) = |\alpha|^2, \quad P(1) = |\beta|^2,$$

and

$$|\psi'\rangle = U|\psi\rangle.$$

You should also know why unitary gates preserve normalization and why $H^2 = I$.

Suggested references

- IBM Quantum Learning, *Basics of Quantum Information* and current Qiskit learning material: <https://quantum.cloud.ibm.com/learning/en/courses>
- IBM Quantum Documentation and Qiskit guides: <https://quantum.cloud.ibm.com/docs>
- Quantum Rings, *Quantum 101*: <https://www.quantumrings.com/course/quantum101>
- M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press.

The software-facing details in the practical notebook should be treated as the canonical implementation for this course. The purpose of this booklet is to explain the physics, mathematics, and algorithmic reasoning behind that implementation.